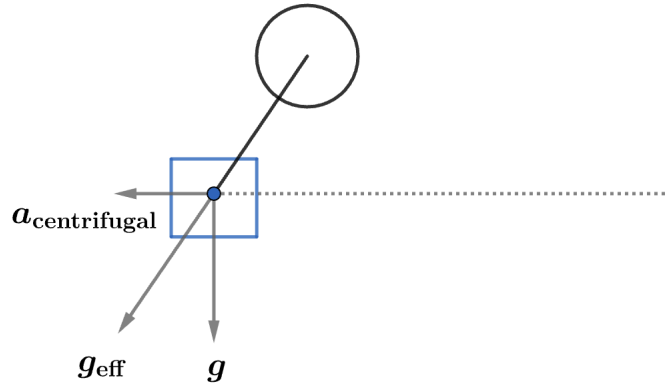


2010 F=ma Exam: Problem 22

Kevin S. Huang



In the noninertial frame of the car, we add a centrifugal force directed outwards from the center of the circle. Combining the fictitious force with gravity,

$$m\vec{g} + m\vec{a}_{\text{centrifugal}} = m\vec{g}_{\text{eff}}$$

We find the effective gravity points in the lower left direction. The balloon floats in the opposite direction so the answer is D.